

Wooah Choi

AI Safety and Evaluation | Machine Learning, NLP and Search

wooah.choi2@gmail.com | +44 7922 651195 | [linkedin.com/in/wooah-choi](https://www.linkedin.com/in/wooah-choi) | github.com/hncpyj
wooahchoi.com | backenddeveloper.tistory.com | London, UK | No visa sponsorship required until 2029

Machine learning engineer working on AI safety and evaluation, with 3+ years across production software engineering and ML research. Built production NLP and search systems over high-stakes legal and financial data for the Supreme Court of Korea, Korea Eximbank and consumer technology, including taking Supreme Court litigation search from 5+ minutes to 8 seconds. MSc in Artificial Intelligence at Queen Mary University of London, with a dissertation on mechanistic interpretability of a PPO-trained Transformer policy in a partially observable reinforcement learning task. Sole author of a technical report on control boundaries for autonomous research agents, showing that a curator model can shift an overseer's decisions while its explicit approval rate stays unchanged. Co-author on a paper accepted to FinNLP 2026 at EMNLP on whether retrieved web evidence is sufficient to verify a financial claim.

Skills

NLP, Search & ML	BERT, DistilBERT, ColBERTv2, Hugging Face Transformers, LLMs, supervised fine-tuning, LoRA, RAG, semantic and vector search, FAISS, BM25, RRF, ranking and recommendation, named entity recognition, query understanding, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, time-series forecasting
Evaluation & Statistics	evaluation framework design, benchmark construction, mechanistic interpretability, fault injection and adversarial stress testing, model validation, causal and counterfactual evaluation, A/B testing, significance testing (z-tests, Welch's t-test), Benjamini-Hochberg correction, effect sizes (Cohen's h and d), bootstrap confidence intervals
Languages & Infrastructure	Python, Java, C++, SQL, TypeScript, R, Bash, Oracle, PostgreSQL, MySQL, Docker, Jenkins, GitLab CI, AWS (EC2, S3, RDS), Linux, Spring, FastAPI, REST APIs, GraphQL

Education

MSc in Artificial Intelligence | Queen Mary University of London, London, UK Sep 2025 – Sep 2026

- **Expecting Distinction**
- MSc dissertation on mechanistic interpretability of a Transformer policy in a partially observable RL task
- Core Modules: Machine Learning, Artificial Intelligence, Natural Language Processing, Reinforcement Learning, Information Retrieval, Neural Networks and NLP, Conversational Agents and Dialogue Systems, *Ethics, Regulation and Law in Digital Decision Making*

BSc in Computer Science | Korea National Open University, Seoul, South Korea Mar 2021 – Aug 2023
Double Major: Statistics and Data Science

- Cumulative GPA 3.3/4.5 (87%), UK 2:1 equivalent
- Core Modules: Multivariate Analysis, Regression Models, Bayesian Data Analysis, Data Mining, Forecasting Methods, Discrete Mathematics (A+), Linear Algebra (A+), Deep Learning (A)

BEng Environmental Engineering | Catholic University of Korea, Seoul Mar 2013 – Feb 2018

Publications and Academic Service

Oct 2026 *Relevant Is Not Enough: Expert-Audited Decision-Grade Web Evidence for Financial Fact Verification.* Park, H., Kim, E., **Choi, W.**, & Zubiaga, A. FinNLP 2026 @ EMNLP 2026, Budapest. **Accepted (poster), to appear.** Contribution: annotated 500 of the 750 claim–passage pairs under a blinded four-class protocol, including the 250-pair double-annotated block behind the paper's Cohen's kappa.

Aug 2026 Reviewer, FinNLP 2026 @ EMNLP 2026 (*ACL SIG-FinTech*)

Work Experience

GridFlow Trade, London | *Co-Founder, AI and Backend* | gridflowtrade.com Nov 2025 – Apr 2026
Co-founded a UK energy trading startup. Selected for the QMUL QIncubator programme and pitched to an investor panel, April 2026.

- Owned the forecasting product end to end: 48 features on 55,000+ half-hourly settlement periods from Elexon and NESO, including price lags, weather and generation cross-features, **Loss of Load Probability** and **de-rated generation margin**. Benchmarked Naive → Ridge → LightGBM → XGBoost → 2-layer LSTM under time-based splits, with XGBoost quantile regression reaching MAE £4.10/MWh ($R^2=0.943$) against £17.13 naive, a **76% error reduction**
- Deployed the quantile model as a production decision layer, issuing charge, discharge and hold actions across 48 daily settlement periods under price uncertainty, and set the commercial case for battery storage trading on the UK Balancing Mechanism

Konan Technology Inc, Seoul | *Software Engineer, AI Solution Development Team* Nov 2020 – Jun 2023
Production search and NLP for more than four enterprise clients across banking, insurance, the judiciary and consumer tech, plus fourteen shorter projects. Worked with sensitive legal and financial records where retrieval quality, access control and system availability were operationally consequential. Owned the full lifecycle from requirements and architecture through to deployment and handover.

- **Supreme Court of Korea** | Next-generation electronic litigation search and analysis [Java, Spring, Jenkins, GitLab, Python, Linux, Docker, C++, PyTorch, BERT, Oracle]
 - Built the query understanding layer: fine-tuned BERT-based NER over case numbers, statutes and party names, a **4%+**

gain in entity extraction, with speech act analysis and semantic role labelling for intent. Entities typed at index time so queries hit the right field, not the whole corpus

- Rebuilt the Java backend and indexing pipeline: search latency **5+ minutes to 8 seconds**, 375+ misanalysed cases resolved, facet search 9.8s to 0.06s
- Semantic and vector search for query expansion and typo correction, related and recommended keywords via LDA on search logs, Jenkins CI/CD on GitLab, and search analytics dashboards for administrators
- **Korea Eximbank** | Records management system, integrated search with high availability [Java, Oracle, Linux]
 - Dual-server search architecture for a state policy bank: single-writer indexing with forced failover, index replication on recovery and L4 load balancing, so search stayed available without the nodes diverging
 - Took on project management and client negotiation. The specified redundancy topology could not fail over, so I built the case for a non-technical stakeholder and the design was changed
 - Sized index partitioning against a 4 million record, terabyte-scale corpus, and delivered disclosure-level access control over a four-level classification scheme
- **Millie's Library** | Book search and user pattern analysis [C++, Python, MySQL, Linux]
 - Automated indexing for 100M+ records, with a volume structure designed to scale beyond 1B
 - Built the ranking layer from nothing, combining field match weights with 7 and 30-day engagement signals. **Ranking quality improved 38%** against a fixed test query set
 - Personalised recommended search terms with no labelled data, grouping accounts on search time and estimating term demand from history
- **Heungkuk Fire & Marine Insurance** | Main website integrated search renewal [Java, JavaScript, Oracle, Linux]
 - Rebuilt the integrated search for a major Korean insurer, full stack from indexing to the customer-facing search page, with auto-completion and recommended search terms driven by customer query logs
- **Internal enablement** | Trainer for other engineers on the search engine and standard source code, producing the technical documentation and user guides

Projects

Research Agent Lab | github.com/hncpyj/research-agent-lab

2026 – Present

Protocol Before Code: Stress-Testing Control Boundaries in Autonomous Research Agents. Choi, W. Technical report, 2026.

- Built the research agent and designed *Protocol Before Code*, a defense-in-depth control architecture against **silent scientific objective drift**, where a pipeline runs but no longer tests the human-approved study. Controls include a machine-checkable StudyProtocol frozen by hash, separate intent-fidelity and methodology gates, content-hash binding of trusted components, least-authority execution tiers, and independent validation of scientific conformance and software correctness.
- Built fault-injection, holdout, and automated evaluation frameworks for agent-control boundaries. **0/18** drifted protocols passed intent-fidelity checks and **18/18** behavior-changing trusted-component mutations were blocked before execution. Disabling content-integrity detection allowed **15/18** identical mutations to escape the remaining stack, isolating the safeguard's marginal value; 6/6 behavior-preserving edits were also refused, quantifying its strictness cost.
- Ran a preregistered **192-pair** LLM oversight experiment where a curator model selected options shown to an overseer model. Adversarial curation increased designated-target selection from **13.0% to 27.1%** (+14.1 pp; predeclared paired bootstrap CI 8.3–20.3) while explicit approval remained **100%** in both conditions, so the decision shifted with no change in the oversight signal. Used preregistered metrics, append-only records, and claim-to-evidence auditing for reproducible safety evaluation.

MSc Dissertation | QMUL · Supervisor: Paulo Rauber

Feb 2026 – Sep 2026

Causal Cue Retrieval and Greedy Route Execution: A Mechanistic Analysis of a Transformer Policy in a Fixed MiniGrid T-Maze

- Trained and evaluated causal Transformer, parameter-matched LSTM, and MLP policies with PPO in a partially observable MiniGrid task, using cue-stratified metrics to distinguish genuine memory use from shortcut policies hidden by aggregate success.
- Developed mechanistic interpretability audits using cue ablation, retrieval blocking, opposite-cue memory patching, and norm-preserving source swaps; showed that correlational attribution could misidentify the internal source causally controlling a policy decision.
- Diagnosed policy failures across seeds, checkpoints, and action-selection rules: sampling recovered 4/7 apparent argmax failures to $S_{\min} = 0.970\text{--}1.000$, and the clean mechanism signature held in only 3 of 6 checkpoints that solved the task equally well, so equal behaviour did not imply equal mechanism across retrains.

Multimodal VQA and LLM Fine-Tuning

2026

MSc coursework, QMUL · Neural Networks and NLP; Conversational Agents and Dialogue Systems

- Built a LLaVA-style multimodal model from scratch: frozen LongCLIP-ViT-B/32 vision encoder, Qwen3-0.6B language model and a trainable projector, with LoRA added in a second phase. Caption restoration 99.83%, base VQA 61.36%, and **77.85%** final after caption-aware late fusion, seed ensembling and option-shuffle TTA
- Compared full supervised fine-tuning against LoRA on Gemma-3-270M: LoRA reached F1 **33.00%** against 32.16% for full SFT while training **2.75%** of parameters. Also fine-tuned Qwen3.5-0.8B with LoRA, raising task accuracy from 10% to 20%

Multi-Domain Neural Search Engine and RAG Grounding Evaluation

Feb 2026 – Sep 2026

MSc Information Retrieval coursework, QMUL · Team project; grounding and statistics added independently

- Built a 6-domain search engine: domain-specialised BERT encoders (SciBERT, BioBERT, **FinBERT**, LegalBERT, ClinicalBERT), a fine-tuned DistilBERT router, hybrid BM25 + FAISS-HNSW with RRF fusion and ColBERTv2 reranking
- Added a RAG grounding framework: 6 causal context-perturbation conditions measuring whether answers depend on the retrieved context or on parametric memory
- Built the A/B testing infrastructure: two-proportion z-test, paired Welch's t-test, Benjamini-Hochberg FDR correction and Cohen's h/d effect sizes across 21 metrics, validated with a 5-step ablation ladder

[Kaggle] Google AI4Code, Understand Code in Python Notebooks

May 2022 – Aug 2022

- Predicting the next cell to be executed from the comments and code in Python Jupyter notebooks; ensemble of CodeBERT and DistilBERT weighted 0.748 / 0.252 on measured performance. **Top 17%** on the private leaderboard